

NETHEESWARAN A

DATA SCIENTIST | PRODUCT ANALYTICS & MACHINE LEARNING

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Professional Summary

Data Scientist with 4 years of experience applying machine learning and analytics to drive business and product decisions. Skilled in statistical modeling, experimentation, and analyzing user behavior to generate actionable insights. Experienced in working with large-scale datasets, building data pipelines, and collaborating with stakeholders to improve product performance and value delivery.

Technical Skills

- **GenAI Engineering:** Multi-Agent Systems, RAG Architectures, ReAct/CoT/Self-Ask Patterns, Prompt Engineering, Guardrails & Safety.
 - **LLM Providers:** OpenAI, Anthropic, Google Gemini, Hugging Face (Llama/Mistral)
 - **Backend Engineering:** Python, FastAPI, REST APIs, Microservices
 - **Machine Learning:** NLP, Embeddings, Fine-Tuning, Model Evaluation, Hugging Face Transformers
 - **DevOps & Infra:** Docker, Kubernetes, GitHub Actions (CI/CD), Monitoring/Logging (MLflow, LangSmith)
 - **Workflows:** Airflow, Prefect
 - **Data Engineering:** SQL, PostgreSQL, Snowflake, Data Lakes, ETL/ELT Pipelines
 - **Cloud:** AWS (EC2, S3), Azure Synapse/Data Lake, GCP (basic)
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Professional Experience

TATA Consultancy Services (TCS) | Data Scientist | Bangalore | Mar 2022 – Aug 2024

Responsibilities:

- Built machine learning models to analyze large scale datasets (10M+ rows), identifying patterns and trends to support business decision-making.
- Developed statistical models and performed exploratory data analysis to uncover drivers of performance and improve prioritisation accuracy by 18%.
- Designed data pipelines and preprocessing workflows, improving data quality and reducing errors by 30% for downstream analytics and modeling.
- Analyzed structured and unstructured data to generate insights on user behavior and system performance, supporting data-driven product improvements.
- Built scalable data pipelines for training, evaluation, and batch inference, processing 100K+ records per run in production environments.
- Developed dashboards and visualizations in Power BI and Python to communicate insights and support stakeholder decision making.
- Collaborated with cross-functional stakeholders to define metrics, translate business requirements, and deliver actionable insights.

Projects:

Predictive Financial Risk & Forecasting Model

- Built a supervised ML model to analyze and prioritise 500K+ records, improving accuracy by 18% and supporting data-driven decision-making.
- Engineered features from large-scale Snowflake datasets (10M+ rows), identifying patterns and trends to improve model performance.
- Designed data validation workflows, reducing data errors by 30% and ensuring reliable inputs for analysis and modeling.
- Deployed the model on Azure with MLflow tracking and CI/CD pipelines, enabling scalable batch processing (100K+ records/run).

PaaS Insurance RAG Application

- Developed an AI-driven system to analyze unstructured documents, enabling users to extract insights from policy, claims, and regulatory data.
- Built data pipelines to process and structure unstructured text, improving accessibility and usability of information.
- Optimized retrieval and ranking logic, reducing hallucinations by 40% and improving output reliability.
- Evaluated system performance using retrieval and language model metrics to ensure accuracy and consistency.

LLM Fine-Tuning for Investment Question Answering

- Developed a domain-specific model to improve question answering accuracy on investment and policy datasets.
- Created structured datasets from historical data to enhance model performance and domain understanding.
- Evaluated model performance using standard NLP metrics and benchmark comparisons.
- Integrated the model into a retrieval-based system, improving response accuracy by 30% for complex queries.

Cloud Data Analyst - Snowflake | Bangalore | Nov 2020 – Mar 2022

- Performed validation for Azure Teradata to Snowflake migration (12TB, 420+ tables), applying reconciliation frameworks and advanced SQL to ensure reliable datasets for downstream machine learning systems.
- Built an automated validation suite of 680 SQL/Python tests (pytest, ADF, GitHub Actions, Slack/email alerts), improving defect detection by 97% and reducing QA cycle time by 48 hours per release.
- Designed Power BI monitoring dashboard for data quality KPIs, improving transparency and delivering actionable insights for business stakeholders through data visualization
- Authored comprehensive documentation and governance playbooks for quantitative analytics workflows, reducing MTTR by 35% and strengthening audit readiness.

Personal Projects

AI Medical Assistant - Tech stack - Python, Pinecone, OpenAI, Langchain, Groq, HuggingFace, Flask - [GitHub](#)

- Developed an AI Medical chatbot that answers medical queries by loading medical PDF into the Pinecone vector database and utilising Groq for NLP.
- Deployed on AWS EC2 with GitHub Actions for CI/CD.

AI Career Helper - Tech stack - Python, FAISS, OpenAI, LangChain, Streamlit - [GitHub](#)

- Designed and developed an AI-powered assistant to guide users on careers in AI, Machine Learning, and Data Science.
- The system extracts and embeds content from industry career guides and interview prep documents into a FAISS vector index, enabling semantic search and context-aware query resolution.

Education

Masters in Data and Computational Science | Sep 2024 - Aug 2025 | Dublin, Ireland

University College Dublin - Grade 2:1 - 3.6/4.2

Bachelors in Electronics and Communication Engineering | July 2016 - Mar 2020 | Erode, India

Kongu Engineering College - CGPA: 8.0/10

Certifications & Courses

MLOps (Coursera) | Data Science Specialization (Coursera) | Python for ML & Data Science (Udemy) |

Snowflake Complete Masterclass (Udemy) | SQL Advanced (Hackerrank) | Microsoft Power BI (Udemy)